

Prior accumulator models in unified notation

Translating McMurray, Mitchell–McMurray, Hidaka, Mollica–Piantadosi,
and Kachergis into our $(\alpha_i, r_i, \kappa_i, \delta_j)$ coordinates

Standard Model 2 working note

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1 Why this exercise

Each prior accumulator model uses its own notation: McMurray writes a threshold density $g(\tau)$, Hidaka uses rate-change exponent D , Mollica & Piantadosi use a per-word (k_w, λ_w) , and so on. Translating each model into our common parameterization makes the comparison literal rather than structural. It also pins down exactly which of our parameters each prior model pins (and at what value), versus what they leave free.

The unified parameterization throughout this note is

$$\eta_{ij}(t) = \underbrace{\log r_i}_{\text{input rate}} + \underbrace{\log \alpha_i}_{\text{efficiency}} + \underbrace{\kappa_i \log t}_{\text{scaling}} - \underbrace{\delta_j}_{\text{word difficulty}}, \quad P(\text{produces}_{ij} \mid t) = \sigma(\eta_{ij}). \quad (1)$$

A pure accumulator pins $\alpha_i \equiv 1$ and $\kappa_i \equiv 1$, leaving only r_i and δ_j free. Our model frees both α_i and κ_i at the per-child level.

Each section below: the original model, in its native notation; the translation; what’s pinned vs. free in our coordinates.

2 McMurray (2007): pure accumulator

Original. Each word w has a fixed acquisition threshold τ_w , drawn from a population distribution $g(\tau)$. At each unit time step every word’s accumulator gains one count. Word w is learned at the first t such that the count reaches τ_w . Since accumulation is unit-rate and deterministic, that’s the first $t \geq \tau_w$. Aggregate vocabulary is $L(t) = \int_0^t g(\tau) d\tau = G(t)$ (the CDF of g).

Translation. A deterministic step $\mathbb{I}[t \geq \tau_w]$ is the zero-noise limit of $\sigma(\lambda(t - \tau_w))$ as $\lambda \rightarrow \infty$. With our IRT logit link,

$$P(\text{produces}_w \mid t) = \sigma(\log t - \log \tau_w) = \sigma(\kappa \log t - \delta_w) \quad \text{with} \quad \kappa \equiv 1, \quad \delta_w := \log \tau_w. \quad (2)$$

The threshold density $g(\tau)$ becomes the prior over the per-word δ_j on a log-time scale.

In our parameterization.

Parameter	Setting in McMurray	Free in our model?
$\log r_i$	$\equiv 0$ (constant rate, no per-child variation)	free
$\log \alpha_i$	$\equiv 0$ (no efficiency variation)	free
κ_i	$\equiv 1$ (unit-rate accumulator)	free
δ_j	$= \log \tau_j$, drawn from g	free

Acceleration in this model. The aggregate vocabulary curve $L(t) = G(t)$ is the CDF of g . Whether the population-mean trajectory shows visible acceleration depends entirely on the shape of g :

- If $g(\tau)$ is uniform on a window, $L(t)$ is linear in t over that window.
- If $g'(\tau) > 0$ over the relevant range (more words have higher thresholds than lower), $L(t)$ accelerates.
- If $g'(\tau) < 0$, $L(t)$ decelerates.

McMurray’s argument: empirical $L(t)$ accelerates, so $g'(\tau) > 0$. **This is a population-level claim, not a per-child claim.** McMurray has no κ_i heterogeneity ($\sigma_\kappa = 0$); the “acceleration” he refers to is in g , which corresponds to heterogeneity in our δ_j across words.

3 Mitchell & McMurray (2009): McMurray + leverage

Original. McMurray (2007) plus a feedback term: as the child accumulates vocabulary, each known word “leverages” the acquisition of new words. The effective time becomes

$$a(t) = t + cL(t), \quad L(t) = G(a(t)), \quad \frac{dL}{dt} = \frac{g(t + cL)}{1 - c g(t + cL)}, \quad (3)$$

where c is the leverage strength.

Translation. The leverage $cL(t)$ couples each child’s effective time to their own vocabulary state. In our linear predictor, this is a non-linear-in- L modification:

$$\eta_{ij}(t) = \log a(t) - \delta_j = \log(t + cL_i(t)) - \delta_j. \quad (4)$$

The argument is no longer linear in $\log t$, and $a(t)$ depends on the child’s own vocabulary state, which makes per-(child, word) likelihoods non-separable in our framework. **This is the one prior model that doesn’t embed cleanly in equation (1).**

Mitchell & McMurray’s result. They prove the leverage *shifts* the growth curve forward or backward in time but *cannot create acceleration not already present in $g(\tau)$* . In our coordinates: leverage moves the timing of acquisition without changing κ_i — it can’t manufacture super-linear scaling.

4 Hidaka (2013): closed-form AoA families

Original. Three nested learning processes, each with closed-form AoA distribution:

Process	Dynamics	AoA distribution
Cumulative	N -accumulator at constant rate δ	$\Gamma(N, \delta)$
Rate-change	1-accumulator at rate $\delta \cdot t^D$	Weibull
Cumulative + rate-change	both	Weibull–Gamma

The rate-change form yields ability $\propto t^{D+1}$, since $\int_0^t \delta \tau^D d\tau = \frac{\delta}{D+1} t^{D+1}$.

Translation: rate-change form. On the logit scale,

$$\log(\text{ability at } t) = (D + 1) \log t + \text{const}, \quad (5)$$

so equation (1) reduces to

$$\eta_{ij}(t) = (D + 1) \log t - \delta_j \quad \text{with} \quad \kappa_i \equiv D + 1. \quad (6)$$

Hidaka’s D is exactly $\kappa - 1$ in our notation. His $D > 0$ is our $\kappa_{\text{pop}} > 1$ (super-linear scaling). Crucially, D is shared across all kids — no per-child variation.

Translation: cumulative form. The Gamma waiting-time form $\Gamma(N_w, \delta)$ does not have a simple closed-form on the log scale; for large N_w it converges to a normal CDF, which approximates a sigmoid. Mapping at the mean: $\delta_j \approx \log(N_j/\delta)$, and $\kappa_i \equiv 1$. This is essentially McMurray’s pure accumulator with a smoothed threshold.

In our parameterization (rate-change).

Parameter	Setting in Hidaka rate-change	Free in our model?
$\log r_i$	$\equiv \log \delta_{\text{const}}$ (population rate)	free
$\log \alpha_i$	$\equiv 0$ (no efficiency variation)	free
κ_i	$\equiv D + 1$ (shared, no per-child variation)	free
δ_j	per-word, free	free

Hidaka’s posit: D is informative about word properties. He reports correlation between D and imageability, taking D from population-pooled fits. In our coordinates this corresponds to per-word κ_w (not per-child κ_i), which we don’t currently estimate. (It’s an interesting direction.)

5 Mollica & Piantadosi (2017): per-word Gamma waiting time

Original. Per-word generative model: word w acquisition begins at start age s_w , “effective learning instances” (ELIs) arrive as Poisson at rate λ_w , and the word is acquired after k_w ELIs. So

$$\text{AoA}_w \sim s_w + \Gamma(k_w, \lambda_w), \quad F_w(a) = \int_0^{a-s_w} \frac{u^{k_w-1} e^{-u\lambda_w} \lambda_w^{k_w}}{\Gamma(k_w)} du. \quad (7)$$

Mollica & Piantadosi report $k \approx 10$, $\lambda \approx 0.5/\text{mo}$, $s \approx 2$ mo for comprehension across 13 languages.

Translation. For large k_w , the Gamma CDF converges to a Gaussian CDF (CLT) with mean k_w/λ_w and standard deviation $\sqrt{k_w}/\lambda_w$. The Gaussian CDF approximates the logistic with a ~ 1.7 scale factor. So

$$F_w(a) \approx \sigma(b_w(a - k_w/\lambda_w)), \quad b_w \propto \lambda_w/\sqrt{k_w}. \quad (8)$$

On the log scale (writing $a = e^{\log t}$, no s_w for simplicity), the curve is approximately a sigmoid in $\log t$ with slope proportional to $b_w \cdot k_w/\lambda_w = \sqrt{k_w}$.

In our parameterization:

$$\eta_{ij}(t) = \log t - \delta_j \quad \text{with} \quad \delta_j \approx \log(k_j/\lambda_j), \quad \kappa_i \equiv 1. \quad (9)$$

κ doesn't appear directly because the Gamma waiting time has a fixed shape; the per-word slope variation comes from $\sqrt{k_w}$ inside the sigmoid scale, which our model doesn't have a free parameter for (we'd need a per-word discrimination λ_j , the 2PL extension).

In our parameterization.

Parameter	Setting in Mollica & Piantadosi	Free in our model?
$\log r_i$	$\equiv \log \lambda$ (shared rate)	free
$\log \alpha_i$	$\equiv 0$ (no per-child variation)	free
κ_i	$\equiv 1$ (unit-rate accumulator)	free
δ_j	$= \log(k_j/\lambda_j)$, free	free

6 Kachergis et al. (2021): 1-PL Rasch IRT with covariates

Original.

$$\text{logit } P(y_{ij} = 1) = \theta_i + d_j + \beta_a \cdot a_i + \beta_t \cdot \log(p_j H a_i), \quad (10)$$

where θ_i is per-child latent ability, d_j is per-word difficulty, a_i is age, p_j is the word's CHILDES probability, H is waking hours per month, and β_a, β_t are shared slopes (no per-child variation).

Translation. Identify $\theta_i = \log r_i + \log \alpha_i$ (their per-child intercept), $d_j = -\delta_j$ (their per-word intercept), and the linear age covariate $\beta_a a_i$ as a linearization of $\kappa \log a$:

$$\eta_{ij} \approx (\log r_i + \log \alpha_i) + \log p_j + \log H + \kappa \log a_i - \delta_j, \quad (11)$$

with $\kappa \equiv \beta_a / (\partial \log a / \partial a) \cdot a$ a constant population slope, no per-child variation.

In our parameterization.

Parameter	Setting in Kachergis 2021	Free in our model?
$\log r_i$	free (population μ_r , no per-child variance specified)	free, σ_r pinned
$\log \alpha_i$	free (their θ_i absorbs $\log r + \log \alpha$)	free
κ_i	$\equiv$ shared population slope (no per-child variance)	free
δ_j	free (their d_j)	free

Kachergis et al. are the first prior model to free σ_α . But σ_κ is still zero: every child has the same acceleration slope.

7 Unified table

In our coordinates, every prior model pins or frees specific parameters. The pattern below makes clear what’s structural and what’s notational across the literature.

Model	σ_α	κ_{pop}	σ_κ	Per-word δ_j ?
McMurray (2007)	0	1	0	free, $\delta_j = \log \tau_j$
Mitchell & McMurray (2009)	0	1*	0	free, plus leverage c
Hidaka (2013) cumulative	0	1	0	free, $\delta_j = \log(N_j/\delta)$
Hidaka (2013) rate-change	0	$D+1$	0	free
Mollica & Piantadosi (2017)	0	1	0	free, $\delta_j = \log(k_j/\lambda_j)$
Kachergis et al. (2021)	free	free	0	free
Best-fitting model (this work)	free	free	free	free

*Mitchell & McMurray’s leverage is not linear-in-log t , so κ as defined in our model isn’t quite the right summary. Their model lives outside our power-law family.

8 What’s structurally new in our model

Reading down the table, the prior literature progressively frees parameters: McMurray gives all four to a single shared form; Hidaka’s rate-change frees κ_{pop} ; Kachergis frees σ_α . Each step adds one degree of freedom.

Our model is the first to free σ_κ : per-child variation in the scaling exponent. That’s the load-bearing addition that enables fitting the variability signature jointly with acceleration. The 4-panel comparison (deck slide 13) shows visually that this addition is necessary — variants without σ_κ free can’t reproduce both the percentile fan width and its growth with age.

9 Discussion: what this implies for the literature

Acceleration was always in the field, but as a population-level claim. McMurray’s $g'(\tau) > 0$ argument and Hidaka’s $D > 0$ both posit acceleration at the population level. We replicate that claim ($\kappa_{\text{pop}} \approx 10.4 \gg 1$ in English M.best). What’s new is per-child variance: kids differ substantially in their scaling exponent.

Variability was treated as a measurement issue rather than a structural feature. Kachergis’s θ_i first frees the intercept, but the slope is shared. Treating between-child variance as primarily intercept-level is consistent with much of the field’s focus on “per-child ability” as a single number. The data argue for a richer two-axis structure (intercept + slope variance).

The per-child slope variance is a falsifiable claim, not a parameter convenience. If σ_κ were small, our model would collapse onto Kachergis (2021). The empirical posterior gives $\sigma_\kappa = 3.47$ [3.14, 3.84] in English, well above zero and large compared to $\sigma_\alpha = 1.71$. This is the quantitative finding that requires explanation.